1. Methods

1.1. Data Sources and Integration

Substance records were compiled from multiple regulatory lists published through the European Chemicals Agency (ECHA), including the candidate list of substances of very high concern (SVHC), the authorisation list, the restriction list, the persistent organic pollutants (POPs) list, the harmonised classification and labelling list, the EU positive list, and the SVHC identification list, as well as the EU Pesticides Database and the Flemish administrative *sommatie_stoffen* list. For each substance record, the available identifiers were collected: CAS registry number, EC number, and substance name.

Where a CAS number was present, the corresponding InChIKey was retrieved via the PubChem PUG-REST API [1]; where a CAS lookup yielded no result, a fallback lookup by substance name was attempted. All API responses were cached locally using the MD5 hash of the query as the filename, to avoid redundant network requests. A delay of 0.2 s was enforced between successive requests to comply with the PubChem rate limit. Known InChIKeys were subsequently propagated across all records sharing the same substance name within the dataset. The resulting dataset was stored as a unified R [2] data frame linking each regulatory entry to its source list and all available identifiers (`data/processed/all_substances.rds`). RDF serialisation and graph merging were performed using Apache Jena 5.6.0.

1.2. Chemical Identity Classification

To characterise the heterogeneity of regulatory substance entries, a rule-based classification scheme was applied to assign each unique substance name to one of seven mutually exclusive entity types. Classification was implemented as an ordered conditional lookup in which the *Molecule* type was evaluated first, so that the availability of a structure-based identifier always takes precedence over textual pattern matching. The seven types are:

1. *Molecule* — any entry for which a valid InChIKey is available, indicating that the substance can be represented as a defined molecular structure;
2. *Substance group* — entries whose names contain phrases such as “and its compounds”, “and its salts”, “and its esters or isomers”, or group acronyms (e.g. PFAS, PCB, PAH, PBDE, HBCDD);
3. *Analytical parameter* — entries describing measurement quantities rather than individual substances, identifiable by phrases such as “Reaction mass of…”, carbon-range notation (C_n–C_m), or semicolon-separated compound lists;
4. *Mixture* — entries corresponding to complex compositions of unknown or variable structure, including trade-name products, petroleum derivatives (naphtha, kerosine, petrolatum), and creosote;
5. *Regulatory entry* — administrative entries referencing legislative artefacts rather than chemical entities, such as entries formatted as “Entry *N*”, references to EU directives, or annex descriptions;
6. *CAS without structure* — entries with a CAS number for which no InChIKey could be retrieved, indicating either that the substance is not indexed in PubChem or that it represents an incompletely characterised (UVCB) substance;

7. *Unclassified* — entries with neither an InChIKey nor a CAS number and not matching any of the above patterns.

1.3. Linkability Taxonomy

Building on the entity-type classification, a seven-tier linkability taxonomy was derived to express the degree to which a regulatory substance entry can be unambiguously associated with a defined chemical structure. For substance-group entries, the presumed base-compound name was extracted by regular expression and an InChIKey lookup via PubChem was attempted, with a 0.2 s delay between requests, yielding a distinction between groups whose parent compound is structure-resolvable and those for which it is not. Analytical parameters and mixtures were consolidated into a single UVCB/Mixture tier. The taxonomy tiers, in decreasing order of chemical specificity, are: (1) Structure (InChIKey); (2) Group — base compound resolvable; (3) Group — base compound not resolvable; (4) CAS without structure; (5) UVCB / Mixture; (6) Regulatory entry; (7) Unidentified.

This taxonomy was serialised as SKOS Collections in RDF Turtle format, with substance URIs minted as `https://data.omgeving.vlaanderen.be/id/concept/chemical_substance/{md5}`. The classified dataset was saved as `data/processed/all_substances_classified.rds` for use in downstream analyses.

1.4. Identifier Mapping Consistency

For all records possessing both a CAS number and an InChIKey, bidirectional mapping consistency was assessed in two directions. First, the number of distinct InChIKeys associated with each CAS number was computed: a CAS number mapping to more than one InChIKey indicates that the same registry identifier has been assigned to multiple molecular structures, for example due to differing stereochemical representations, hydrates, or salt forms. Second, the number of distinct CAS numbers associated with each InChIKey was computed: an InChIKey associated with multiple CAS numbers reflects the common regulatory practice of assigning separate registry identifiers to the same molecular structure in different trade or regulatory contexts. Both distributions were visualised on a $\log_{10}$ frequency axis [3] to reveal the long tail of the distribution. These analyses motivate the use of InChIKey, rather than CAS number, as the primary linking key throughout all subsequent cross-list analyses.

1.5. Structure Coverage and Linking Ratio

For each regulatory source, the *linking ratio* was defined as the number of unique InChIKeys divided by the total number of regulatory records in that source. A ratio of 1 indicates that every record corresponds to a distinct molecular structure; a ratio below 1 reflects structural redundancy (multiple records per structure), a high non-structure fraction, or both. The linking ratio was visualised as a $\log_{10}$–$\log_{10}$ scatter plot with a 1:1 reference diagonal [3,4], enabling comparison across sources spanning three orders of magnitude in size. This analysis provides the per-source structural density estimate that feeds the workload model described in Section 1.10.

1.6. Cross-List Substance Overlap

Four complementary analyses characterised how structure-defined substances are distributed across regulatory sources.

UpSet plots [5] were used to visualise multi-set intersections of structure-defined substances across all sources with at least ten unique InChIKeys; the threshold of ten was chosen as a display heuristic to suppress near-empty sets that produce unreadable intersection bars. The 30 largest intersections were shown, as the full $2^n - 1$ possible intersections are not interpretable for $n > 5$.

Pairwise Jaccard similarity was computed for all source pairs:

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|}, \tag{1}$$

where A and B denote the sets of unique InChIKeys in two sources, respectively. Only structure-defined entries were included. The resulting similarity matrix was subjected to hierarchical clustering using average linkage on the complementary dissimilarity $1 - J$ [6,7] and visualised as a clustered heatmap; text labels were suppressed for $J < 0.02$ to avoid visual noise from near-zero overlaps.

A second UpSet plot was constructed restricted to the seven principal regulatory obligation lists: the restriction list, candidate list, authorisation list, POPs list, EU positive list, harmonised classification list, and SVHC identification list.

Finally, a force-directed network [8] was constructed in which nodes represent regulatory sources (node size $\propto \log_{10}(\text{list size} + 1)$) and edges connect pairs of sources sharing at least five unique InChIKeys (edge width $\propto \log_{10}(\text{shared InChIKeys} + 1)$). The threshold of five shared InChIKeys suppresses edges representing negligible structural overlap that would otherwise clutter the layout.

1.7. Semantic Embedding and Clustering of Non-Structure Substances

Substance entries that could not be resolved to a chemical structure — excluding administrative regulatory entries — were analysed through unsupervised text embedding and clustering. Each substance name was encoded using the `all-MiniLM-L6-v2` sentence-transformer model [9,10], yielding a 384-dimensional dense vector representation.

The optimal number of clusters was determined by computing the silhouette score [11] for k-means solutions with $k \in \{2, \ldots, 12\}$ (25 random starts, maximum 100 iterations, random seed 42). The optimal solution ($k = 6$) was applied using Lloyd's algorithm [12] to partition the substance names into six semantically coherent groups.

For visualisation, the embeddings were projected to two dimensions using UMAP [13] (15 nearest neighbours, minimum distance 0.1, cosine metric, seed 42).

The six resulting clusters were labelled manually upon visual inspection of the projections: (1) reaction masses; (2) complex substances (UVCB with partial structure); (3) petroleum and coal tar fractions; (4) inorganic compounds and metal salts; (5) trade names, codes and biological materials; (6) polymers, fatty acids and surfactants.

1.8. Chemical Taxonomy Matching

Structure-defined and non-structure substances were mapped to standardised chemical classes using two complementary approaches.

For *non-structure substances*, the cosine similarity [14] was computed between the sentence-transformer embeddings of each substance name and the embeddings of all class labels in the ChemOnt 2.1 ontology [15]. Substance names matching patterns characteristic of reaction masses, petroleum fractions, or UVCB descriptions were excluded prior to matching, as these names do not correspond to a single chemical class. For each remaining substance, the top-3 most similar ChemOnt classes were identified and a composite confidence score was computed as

$$c = 0.6 \cdot s_1 + 0.3 \cdot \Delta_{12} + 0.1 \cdot \Delta_{23}, \tag{2}$$

where s_1 is the cosine similarity of the best-matching class and Δ_{12}, Δ_{23} are the score gaps between the first and second, and second and third candidates, respectively. Retaining three candidates rather than one is necessary to compute both gap metrics. High-quality matches were retained subject to the empirically determined thresholds $s_1 > 0.8$ and $\Delta_{12} > 0.05$, selected by sweeping the primary threshold between 0.3 and 0.9 and inspecting the resulting quality landscape.

For *structure-defined substances*, the direct-parent ChemOnt class was retrieved for each unique InChIKey via the ClassyFire REST API [15], with all API responses cached locally to avoid redundant requests.

1.9. Knowledge Graph Construction

All processed substance data — including the linkability taxonomy (Section 1.3), cluster assignments (Section 1.7), and ClassyFire annotations (Section 1.8) — were serialised to RDF Turtle format. Substance URIs were minted using MD5 hashes of InChIKeys for structure-defined entries, or of substance names otherwise, to ensure stable and opaque identifiers. The linkability taxonomy and cluster assignments were represented as `skos:Collection` instances [16], with `skos:member` triples encoding group-to-base-compound parent relationships. Annotations were recorded using the OA Annotation Ontology [17].

All component graphs were merged with the ChemOnt 2.1 SKOS ontology using Apache Jena 5.6.0 `riot`, with semantic enrichment performed via SPARQL CONSTRUCT queries.

To integrate biological hazard evidence, the ChEBI ontology [15] was linked to the knowledge graph through a four-step pipeline implemented as a companion shell script. First, the ChEBI OWL ontology was downloaded from the European Bioinformatics Institute and converted to RDF Turtle using Jena `riot`. Second, the converted ontology was merged with the project's substance taxonomy graph using Jena. Third, a SPARQL CONSTRUCT query (`exact_match_chebi.rq`) linked project substance concepts to their corresponding ChEBI entities via InChIKey string matching (`skos:exactMatch`), using the ChEBI `chemrof:inchi_key_string` property as the join key. Fourth, a second SPARQL CONSTRUCT query (`chebi_annotaties.rq`) traversed the ChEBI role hierarchy via the `RO:0000087` (*has role*) object property and ma-

terialised each biological role assignment as an OA Annotation triple [17] linking the project substance concept to the corresponding ChEBI role class. Finally, a SPARQL SELECT query with `GROUP_CONCAT` (`chebi_to_table.rq`) exported a flat CSV table (`data/processed/chebi.csv`) containing, per substance, pipe-separated lists of biological activity labels and ECHA list memberships. This table is the primary input for the composite priority scoring described in Section 1.11. The unified knowledge graph was stored at `data/processed/rdf/substances_taxonomy.ttl`.

1.10. Workload Estimation for Manual Curation

Regulatory entries that cannot be resolved to a chemical structure must be related to one another through manual expert assessment. For each pair of non-structure entries, an expert must determine which of four mutually exclusive semantic relations holds: (1) *equivalence* — both entries refer to the same set of substances; (2) *subset* — one entry is fully contained within the other; (3) *overlap* — the entries share some but not all substances; or (4) *disjoint* — the entries share no substances. Only by establishing these relations explicitly can the semantic alignment required for cross-list interoperability be achieved.

Because every unique pair must be evaluated independently, the total number of assessments scales quadratically as $\binom{N}{2} = N(N-1)/2$. At an assumed throughput of 100 pair assessments per person-day, the total workload in person-days is

$$D(N) = \left\lceil \frac{N(N-1)}{200} \right\rceil, \tag{3}$$

where N is the number of non-structure entries under consideration. The ceiling function avoids under-estimating the workload: a fraction of a person-day still requires at least one analyst-day.

1.11. Composite Priority Scoring

To rank regulatory substances by urgency, a composite priority score was constructed by combining two independent evidence dimensions: regulatory list membership and intrinsic biological hazard. The approach is conceptually grounded in the ToxPi (Toxicological Priority Index) framework [18], which integrates heterogeneous evidence streams into a single prioritisation score through transparent, domain-specific weights. The input is the flat CSV table produced by the ChEBI pipeline described in Section 1.9 (`data/processed/chebi.csv`), which contains, per substance, the set of ECHA regulatory lists on which it appears and the set of ChEBI biological role labels assigned to it.

1.11.1. Domain 1: Regulatory List Weights

Each regulatory list was assigned a weight on an ordinal scale of 0 to 5, derived directly from the legal consequences triggered by each instrument under REACH [19], CLP [20], and the POPs Regulation [21]. Table 1 summarises the assignments.

The list score for a given substance is the sum of the weights of all lists on which it appears:

$$\text{list_score} = \sum_i w_{\text{list},i}, \tag{4}$$

where $w_{\text{list},i}$ is the weight of list i and the sum runs over all lists containing the substance. The numerical scale of 0–5 is not prescribed by any regulatory framework; it was chosen to reflect the ordinal REACH hierarchy while remaining interpretable, with the ratio 5:0 (SVHC candidate list : REACH registration) mirroring the qualitative difference in legal consequence.

Table 1. Regulatory list weights used in Domain 1 of the composite priority score. The weight hierarchy reflects the legal consequence of each instrument under REACH (Arts. 57–73), CLP (Annex VI), and the EU POPs Regulation.

Weight	List or instrument	Legal basis
5	Candidate list; SVHC identification	Formal SVHC identification, REACH Arts. 57–59; triggers communication obligations (Art. 33)
4	Restriction list; authorisation list; POPs list	Binding restrictions or use prohibition, REACH Arts. 60–73; elimination obligation, POPs Reg. (EU) 2019/1021
3	Restriction/authorisation/POPs process; PBT assessment; ED assessment	Active regulatory scrutiny; PBT/ED assessments correspond directly to SVHC criteria, REACH Arts. 57(d)–(f) and Annex XIII
2	Harmonised CLP list; CLH process; pesticide approval	Legally binding hazard classifications, CLP Annex VI; positive safety determination required, Reg. (EC) No 1107/2009
1	Substance evaluation; dossier evaluation; EU positive list	Regulatory scrutiny without a final hazard determination
0	REACH registration; anhydrous forms	Baseline administrative requirement; no inherent hazard determination, REACH Title II

1.11.2. Domain 2: Biological Hazard Weights

Biological hazard weights were assigned to ChEBI role labels on a scale of 1 to 3, aligned with the SVHC hazard criteria of REACH Art. 57 and the EU Chemicals Strategy for Sustainability [22]. Only substances with a known InChIKey can receive a ChEBI biological role annotation (approximately 21% of the dataset); substances without a ChEBI match are scored on regulatory list membership alone.

CMR properties (carcinogenic, genotoxic, mutagenic, and teratogenic agents) receive a weight of 3, consistent with their equal legal standing as Tier-1 SVHC criteria under REACH Arts. 57(a)–(c). The ChEBI ontology uses the label *teratogenic agent* (via `RO:0000087`) for reproductive and developmental toxicity, rather than the CLP term "reproductive toxicant"; this distinction must be respected when mapping ChEBI roles to REACH hazard classes. Persistent organic pollutants and endocrine disruptors also receive a weight of 3, corresponding to REACH Arts. 57(d)–(f) and the elevated priority assigned to endocrine disruption by the EU CSS [22].

Neurotoxins, ecotoxins, and confirmed allergens (sensitisation category 1A) receive a weight of 2. Neurotoxicity is identified as a priority hazard class in the EU CSS [22]; ecotoxicity is a PBT-adjacent concern under REACH Annex XIII. Neither property is an explicit SVHC criterion under Art. 57, justifying placement below CMR/POP/ED at weight 2.

Sensitisers (category 1B), antimicrobial agents, and pesticidal activities (herbicide, insecticide, fungicide, acaricide, nematocide) receive a weight of 1, reflecting biologi-

cal interaction with living systems without reaching the threshold of systemic concern associated with higher tiers.

The biological hazard score is the sum of the weights of all ChEBI roles associated with a substance:

$$\text{bio_score} = \sum_j w_{\text{bio},j}, \tag{5}$$

where $w_{\text{bio},j}$ is the weight of biological role j.

1.11.3. Composite Score and Aggregation

The composite priority score combines both domains by additive summation:

$$\text{priority_score} = \text{list_score} + \text{bio_score}. \tag{6}$$

Whereas ToxPi computes a weighted average of normalised domain scores (each rescaled to $[0,1]$) [18], additive summation was preferred here because the two domains are complementary and largely independent: a substance may appear on many regulatory lists but lack a ChEBI InChIKey match, or conversely carry strong biological hazard evidence whilst appearing on few lists. Normalising one domain against the other would suppress the contribution of the domain with the smaller raw value, reducing interpretability. As a consequence the score is unbounded and is interpreted in relative rather than absolute terms.

1.11.4. Visualisations

Seven complementary figures were produced to answer distinct questions about the priority landscape:

1. A ranked bar chart of the top-50 substances by composite score, identifying which substances are most urgent overall.
2. A heatmap of ECHA list membership for the top-50 substances, revealing which regulatory instruments drive each substance's score.
3. An UpSet plot restricted to concern-level lists, showing intersection patterns among the highest-weight instruments.
4. A bar chart of hazard-type prevalence, broken down by evidence domain, illustrating which hazard categories most often contribute to the bio-hazard score.
5. A bubble chart of regulatory breadth (number of lists) against bio-hazard score, partitioned into four quadrants using the median number of list memberships as the horizontal threshold and bio_score > 0 as the vertical threshold. The bio-score threshold was fixed at zero rather than the median because the bio-score distribution is zero-inflated (approximately 79% of substances lack a ChEBI match), which would produce a degenerate two-quadrant split at the median. The quadrants identify: (Q1) broad regulatory coverage and biological hazard evidence — the most urgent substances; (Q2) limited regulatory coverage but biological hazard evidence — potential regulatory gaps; (Q3) broad regulatory coverage but no biological hazard evidence — substances requiring structural identification to enable evidence-based assessment; and (Q4) limited coverage and no biological hazard evidence.

6. An alluvial diagram tracing the flow from hazard type to regulatory instrument, revealing whether the regulatory system has captured each hazard class.
7. A treemap in which tile area represents the number of substances per chemical class and tile colour represents the mean priority score, enabling identification of common but under-prioritised chemical classes.

2. Results

2.1. Data Sources and Integration

Integration of the 18 regulatory sources (see Section 1.1) yielded a unified dataset of 41,813 regulatory records. Each record was linked to its originating source and enriched with all available chemical identifiers (CAS number, EC number, substance name, and InChIKey where resolvable). The combined dataset spans the full breadth of European regulatory instruments, from broadly scoped REACH registration data (*reach_registrations*, 25,200 records) to narrowly focused obligation lists such as the SVHC identification list (273 records) and the POPs process list (12 records). The Flemish *sommatie_stoffen* list contributes a further 1,390 records dominated by substance groups without resolved structures.

2.2. Chemical Identity Classification

The rule-based classifier assigned each of the 41,813 regulatory records to one of seven mutually exclusive entity types. The majority — 26,376 records (63.1%) — are classified as *Molecule*. The next largest categories are *Unclassified* (6,359; 15.2%), *CAS without structure* (4,851; 11.6%), and *Analytical parameter* (3,440; 8.2%). The remaining categories are comparatively small: *Mixture* (685; 1.6%), *Substance group* (86; 0.2%), and *Regulatory entry* (16; 0.04%).

Figure 1 illustrates how the entity type composition varies substantially across individual regulatory sources. Sources rooted in pharmaceutical and pesticide authorisation contexts — ED Assessment (85.5% Molecule), CLH Process (84.2%), and the Candidate List (80.6%) — are dominated by structure-defined entries. In contrast, the *sommatie_stoffen* list (3.9% Molecule) and the Restriction Process (29.4%) contain a high proportion of unclassified or group-type entries, reflecting the broader, less chemically specific scope of those instruments.

2.3. Linkability Taxonomy

Application of the seven-tier linkability taxonomy (Section 1.3) to the full dataset reveals that 20,909 entries (62.5%) are fully addressable via a direct InChIKey (tier 1; Figure 2). A further 60 substance group entries could be partially resolved: 15 entries (<0.1%) belong to tier 2 (group with resolvable base compound) and 45 (0.1%) to tier 3 (group with unresolvable base compound). Together these tiers comprise the structure-addressable portion of the dataset.

The remaining 37.5% of entries are not directly addressable by structure. Of these, 3,865 entries (11.6%) carry a CAS number but no resolvable InChIKey (tier 4); 3,622 (10.8%) are UVCB substances or undefined mixtures (tier 5); 15 entries (<0.1%) are administrative regulatory entries (tier 6); and 4,981 (14.9%) are fully unidentified entries with no actionable identifier (tier 7). This last group — nearly one in six regulatory records — represents the most significant barrier to automated cross-list substance matching.

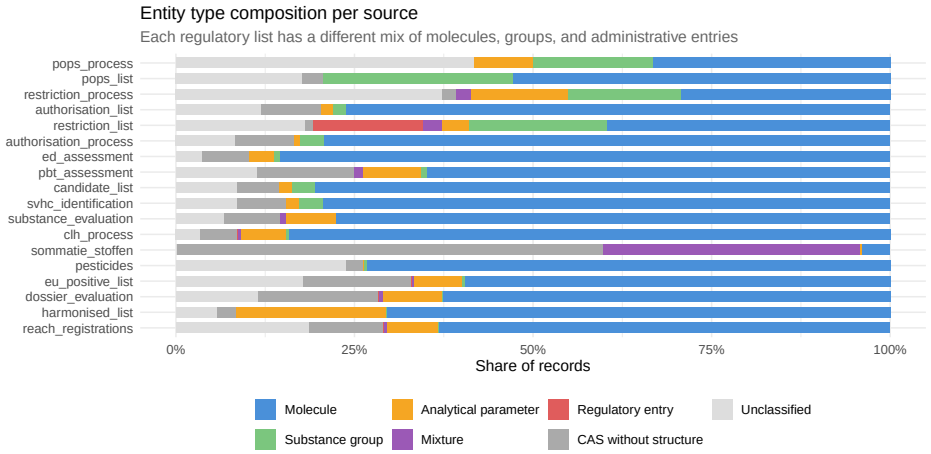

Figure 1. Entity type composition broken down by regulatory source. Sources are ordered by decreasing proportion of Molecule entries. The variation in entity type mix reflects differences in regulatory scope and the specificity of chemical identity requirements.

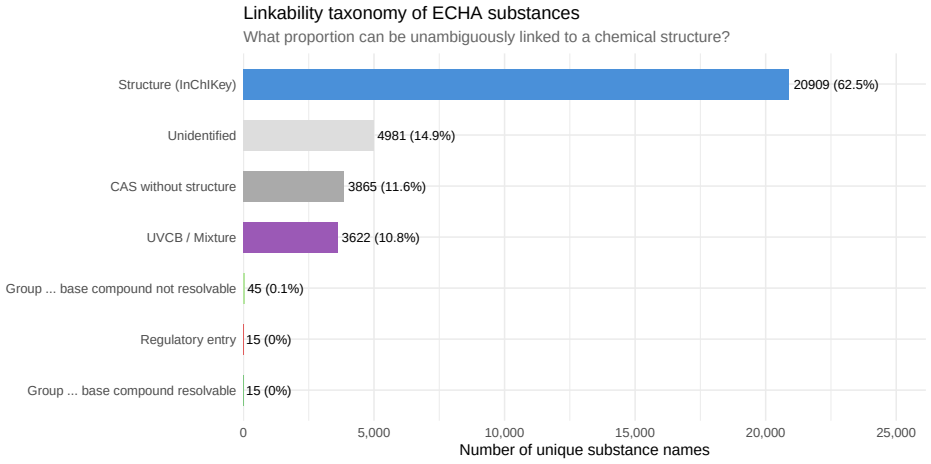

Figure 2. Seven-tier linkability taxonomy applied to all 33,452 unique regulatory substance entries. Tiers 1–3 are structure-addressable (62.5%); tiers 4–7 require alternative or manual approaches to chemical identity resolution.

2.4. Identifier Mapping Consistency

Of the 41,813 regulatory records, 17,277 carry both a CAS number and an InChIKey and could be assessed for bidirectional mapping consistency (Section 1.4).

In the CAS→InChIKey direction, 17,273 of 17,277 assessed CAS numbers (99.98%) map unambiguously to a single InChIKey. The remaining four CAS numbers each map to exactly two distinct InChIKeys. These discrepancies arise from entries in

which the same CAS registry number has been assigned to structurally related but distinct compounds — most commonly stereoisomeric pairs, hydrate/anhydrous forms, or alternative salt representations recorded in different regulatory dossiers.

In the InChIKey→CAS direction, 17,074 unique InChIKeys could be assessed, of which 16,879 (98.9%) are associated with a single CAS number. The remaining 195 InChIKeys (1.1%) each map to two or more CAS numbers (Figure 3), reflecting the common regulatory practice of assigning separate registry numbers to the same molecule under different trade, salt, or regulatory-context identifiers. These findings confirm that CAS numbers alone are insufficient as unambiguous primary identifiers for cross-list substance linking, and motivate the use of InChIKey as the primary key throughout subsequent analyses.

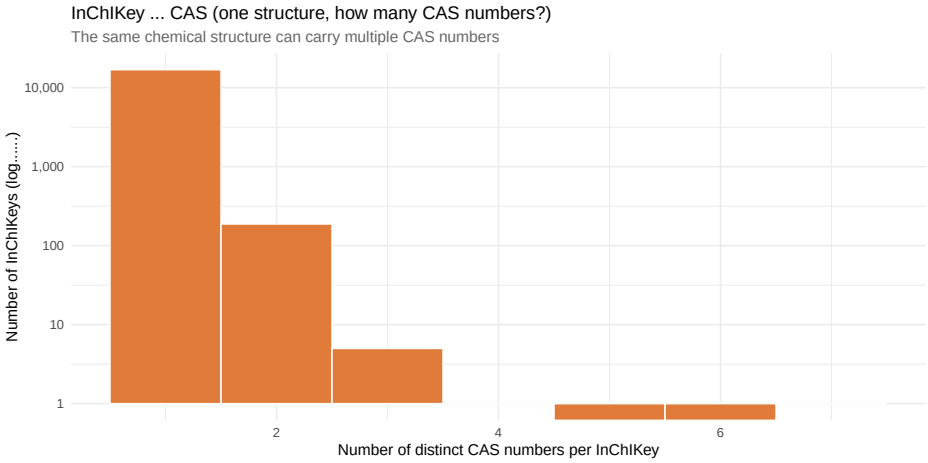

Figure 3. Distribution of the number of distinct CAS numbers per InChIKey across 17,074 assessed structures. One hundred and ninety-five InChIKeys are associated with two or more CAS numbers, indicating registry fragmentation.

2.5. Structure Coverage and Linking Ratio

Across the combined dataset, 26,376 of 41,813 regulatory records (63.1%) are structure-defined, i.e. possess a valid InChIKey. The 20,909 unique InChIKeys thus identified represent the directly linkable chemical space within European regulatory legislation.

Structure coverage varies substantially across the 18 individual sources (Figure 4), ranging from 85.5% (ED Assessment) to 3.9% (*sommatie_stoffen*). Sources operating within tightly defined pharmaceutical or pesticide authorisation frameworks exhibit the highest coverage: ED Assessment (85.5%), CLH Process (84.2%), Candidate List (80.6%), and SVHC Identification List (79.5%). The large REACH Registrations dataset, which contributes 25,200 records — more than 60% of the total — achieves 63.3% coverage. Sources with the lowest coverage include the Restriction Process (29.4%), the POPs Process (33.3%), and *sommatie_stoffen* (3.9%), the latter dominated by substance group entries for which individual molecular structures are not specified.

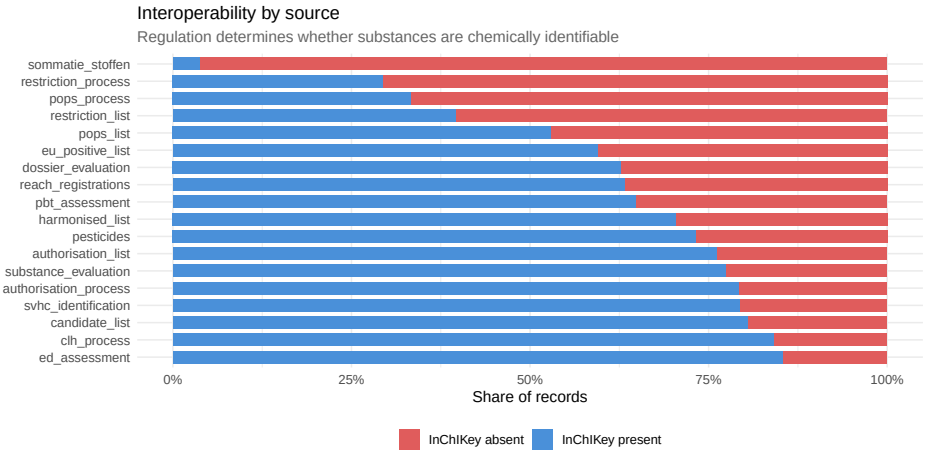

Figure 4. Structure coverage (InChIKey availability) broken down by regulatory source. Sources are ordered by decreasing coverage. The 18-fold range from 85.5% (ED Assessment) to 3.9% (*sommatie_stoffen*) reflects fundamental differences in regulatory scope and the chemical specificity of substance identity requirements.

The log-log scatter plot of records versus unique InChIKeys per source (Figure 5) illustrates the per-source linking ratio. Sources plotting near the 1:1 diagonal (e.g. the POPs list, Candidate List) show a near-one-to-one correspondence between regulatory records and distinct molecular structures. Sources well below the diagonal — notably REACH Registrations, Harmonised Classification List, and Dossier Evaluation — contain multiple records mapping to the same InChIKey, reflecting either systematic structural redundancy (e.g. separate entries for different salt forms of the same compound) or a high fraction of non-structure entries that suppress the effective ratio.

2.6. Cross-List Substance Overlap

The UpSet plot in Figure 6 illustrates the pattern of structure-defined substance overlap across all sources with at least ten unique InChIKeys. The dominant pattern is source-specificity: the largest intersection bars correspond to substances appearing in only a single regulatory list. Shared substances across multiple lists are substantially less frequent, and the shared core across three or more lists is comparatively small. This distribution reflects the domain-specific design of individual regulatory instruments, each targeting a distinct subset of the chemical landscape rather than a common universe of regulated compounds.

The force-directed source network (Figure 7) provides a complementary view, visualising pairwise substance sharing as weighted edges between source nodes. The REACH Registrations hub is the most connected node by virtue of its size, sharing at least five unique InChIKeys with nearly all other sources. Clusters of thematically related sources (e.g. the authorisation-related lists; the CLH and harmonised classification lists) are visually distinct, confirming domain-based structuring of the regulatory chemical space.

The pairwise Jaccard similarity heatmap (Figure 8) further reveals that regulatory sources cluster into domain-specific groups after hierarchical clustering on $1 - J$. The

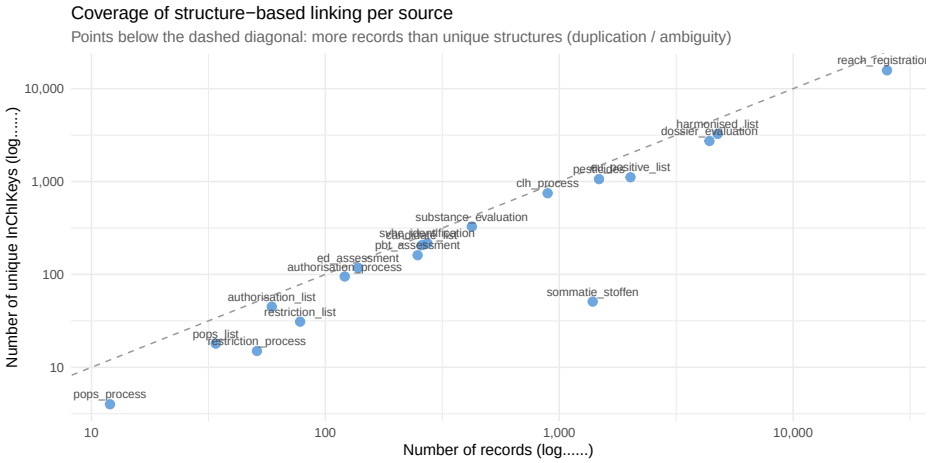

Figure 5. Scatter plot ($\log_{10}$–$\log_{10}$ scale) of the number of regulatory records versus unique InChIKeys per source. The diagonal represents a 1:1 record-to-structure ratio. Sources below the diagonal exhibit structural redundancy or low structure coverage.

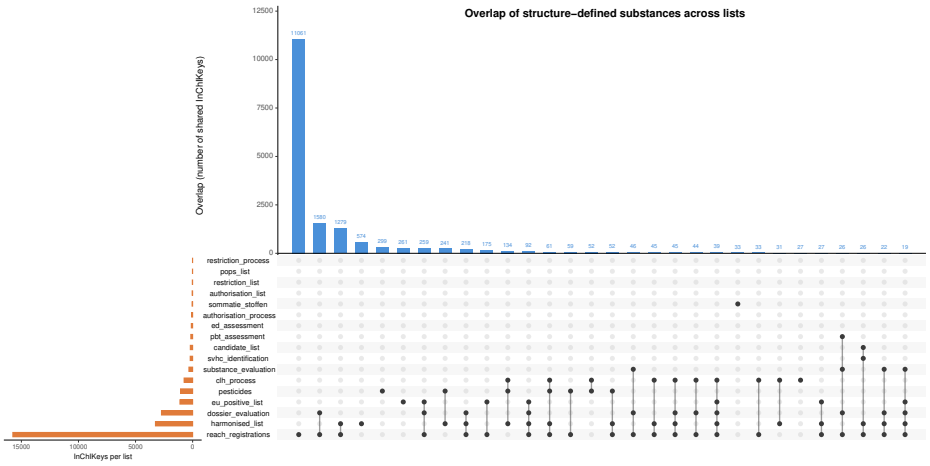

Figure 6. UpSet plot of structure-defined substance overlap across regulatory sources with ≥ 10 unique InChIKeys (top-30 intersections shown). The majority of structure-defined substances appear in a single regulatory source.

SVHC identification list, Candidate List, and Authorisation-related lists form a tight cluster with high mutual Jaccard similarity, consistent with their shared SVHC assessment pathway. The Harmonised Classification List and CLH Process cluster together, reflecting their overlapping scope. The *sommatie_stoffen* list and POPs Process appear as peripheral nodes, sharing little structural overlap with the larger clusters.

Among the seven principal obligation lists, the UpSet plot in Figure 9 shows that the Candidate List and SVHC Identification List share the largest bilateral overlap, as ex-

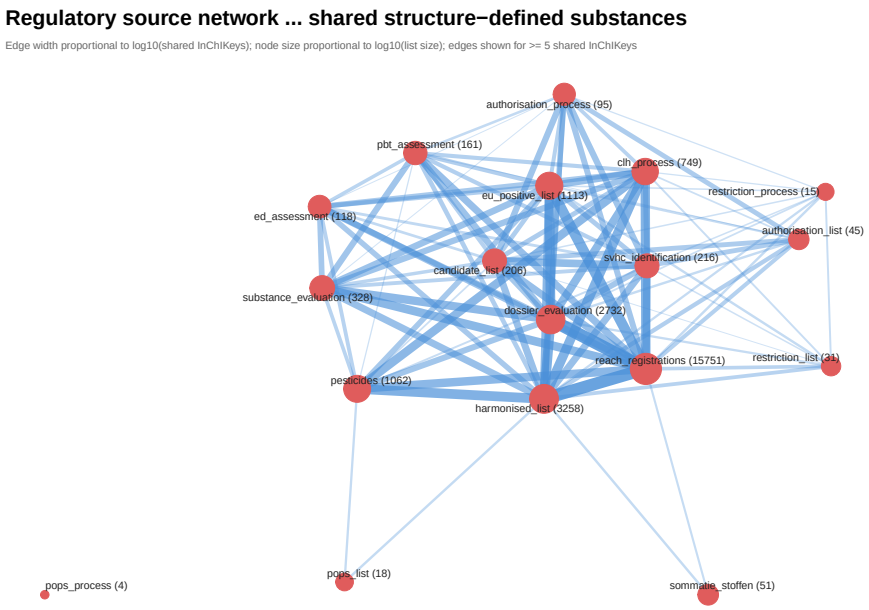

Figure 7. Force-directed network of regulatory sources (Fruchterman–Reingold layout). Node size is proportional to $\log_{10}(\text{list size}+1)$; edges connect source pairs sharing ≥ 5 unique InChIKeys, with edge width proportional to $\log_{10}(\text{shared}+1)$.

pected given their common SVHC assessment pathway. Substances appearing simultaneously in the Restriction List, Authorisation List, and Candidate List form a meaningful intersection representing the most broadly regulated structural substances in European legislation. The POPs list largely stands apart, with limited overlap with the other six lists.

2.7. Semantic Embedding and Clustering of Non-Structure Substances

Silhouette score analysis across $k \in \{2,\ldots,12\}$ (Figure 10) identifies $k = 6$ as the optimal number of k-means clusters for the sentence-transformer embeddings of the 15,432 non-structure substance names (Section 1.7). The silhouette score peaks at $k = 6$ before declining monotonically for larger k, confirming that six clusters provide the best trade-off between cohesion and separation.

The UMAP projection (Figure 11) shows clearly separated cluster regions, confirming that the six groups are semantically coherent. Manual inspection of cluster contents yielded the following labels:

1. *Reaction masses* — defined isomer or intermediate mixtures described by the "Reaction mass of …" naming convention;
2. *Complex UVCB substances* — multi-constituent or variable compositions not reducible to a single structure;

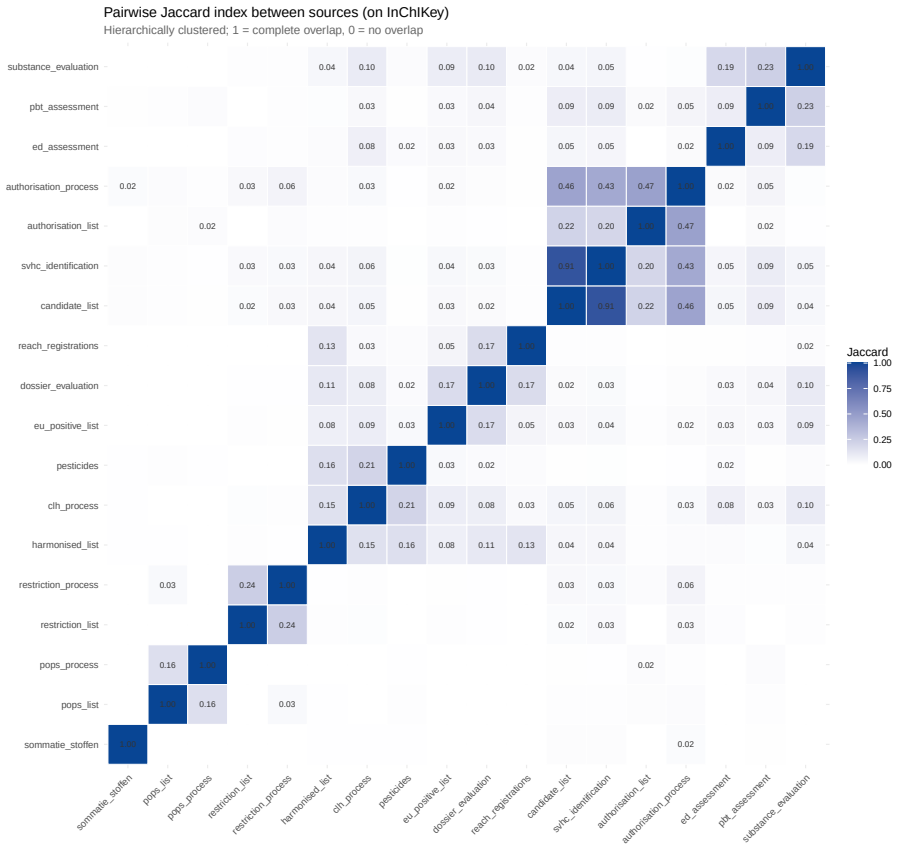

Figure 8. Pairwise Jaccard similarity heatmap of structure-defined substance sets across regulatory sources, ordered by hierarchical clustering (average linkage on $1 - J$). Domain-specific source clusters are clearly visible.

3. *Petroleum and coal tar fractions* — distillation cuts, refinery streams, and tar derivatives;
4. *Inorganic compounds and metal salts* — simple inorganic species, oxides, and coordination compounds;
5. *Trade names, codes, and biological materials* — commercial product identifiers and biological matrix entries;
6. *Polymers, fatty acids, and surfactants* — polymeric substances, long-chain fatty acids, and amphiphilic compounds.

These clusters illustrate that semantic embeddings can organise non-structure substance names into coherent groups, but cannot resolve their chemical identity or establish the formal ontological relationships needed for cross-list interoperability.

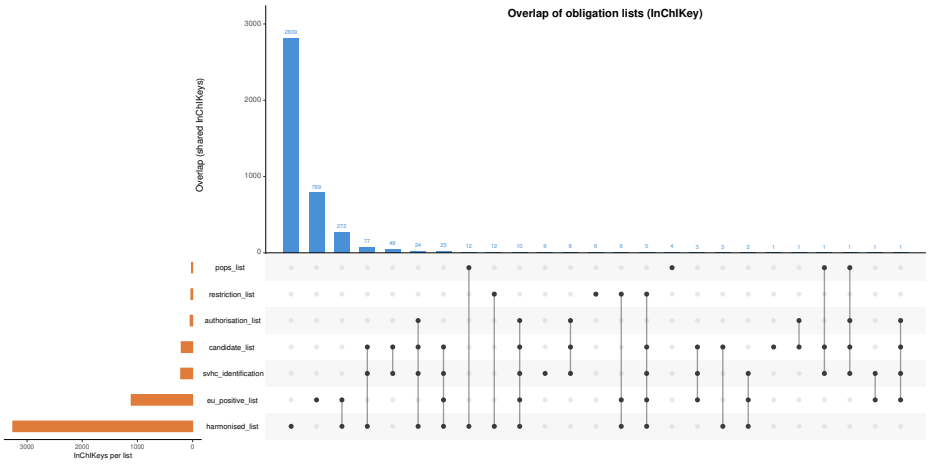

Figure 9. UpSet plot of the seven principal obligation lists (restriction, candidate, authorisation, POPs, EU positive list, harmonised classification, and SVHC identification lists), showing pairwise and multi-list intersection patterns among structure-defined substances.

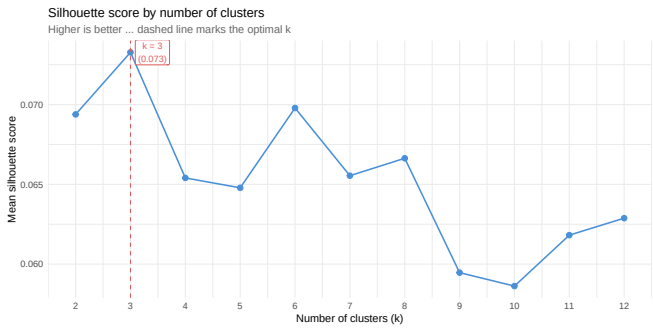

Figure 10. Silhouette score as a function of the number of k-means clusters k for sentence-embedding-based clustering of non-structure substance names. The score peaks at $k = 6$.

2.8. Chemical Taxonomy Matching

2.8.1. Non-Structure Substances

Of the 15,432 non-structure substance names, only 107 (0.7%) received a high-confidence ChemOnt assignment ($s_1 > 0.8$, $\Delta_{12} > 0.05$). The empirical cumulative distribution of best-match scores (Figure 12) shows that approximately half of all non-structure substance names achieve a cosine similarity below 0.5 against any ChemOnt class label, confirming that embedding-based taxonomy matching cannot substitute for structure-based chemical identification for compositional, trade-name, and UVCB-type entries.

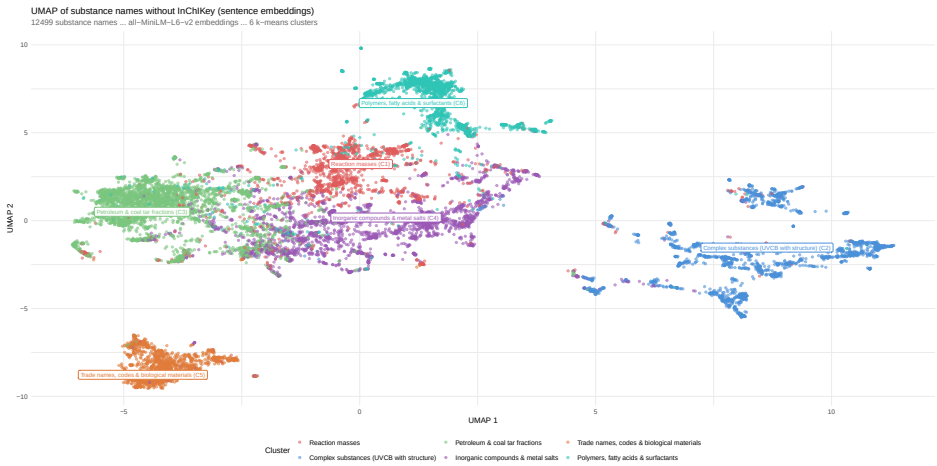

Figure 11. UMAP projection of sentence-transformer embeddings of non-structure substance names, coloured by k-means cluster assignment ($k = 6$). Each cluster corresponds to a semantically coherent chemical or regulatory category.

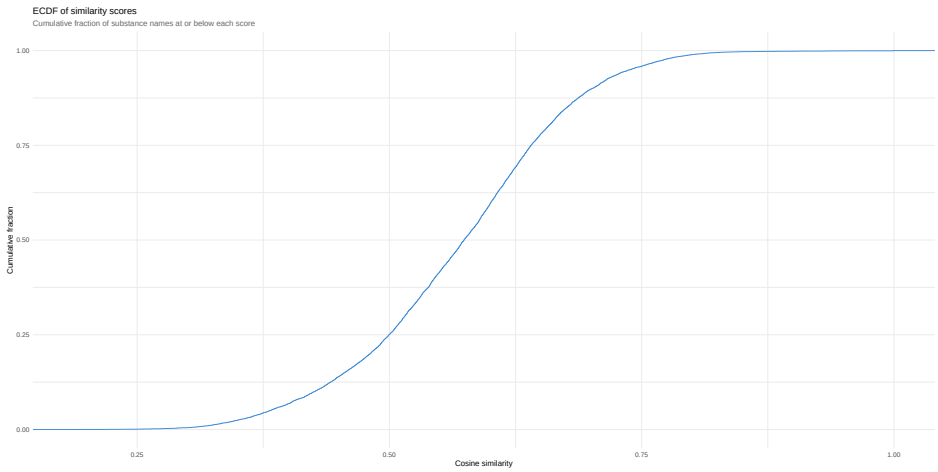

Figure 12. Empirical cumulative distribution function of best-match cosine similarity scores (s_1) for non-structure substance names against ChemOnt class labels. The median match score is below 0.5, reflecting the fundamental mismatch between compositional substance names and structure-based class labels.

2.8.2. Structure-Defined Substances

ClassyFire direct-parent annotations were retrieved via the REST API for the 20,909 unique InChIKeys. Figure 13 presents the top 30 ChemOnt classes by regulatory coverage. Benzenoids form the largest single class, reflecting the prevalence of aromatic compounds across regulatory lists. Organoheterocyclic compounds and carboxylic acids and derivatives are the next most represented classes, consistent with the chemical diversity of industrial substances regulated under REACH.

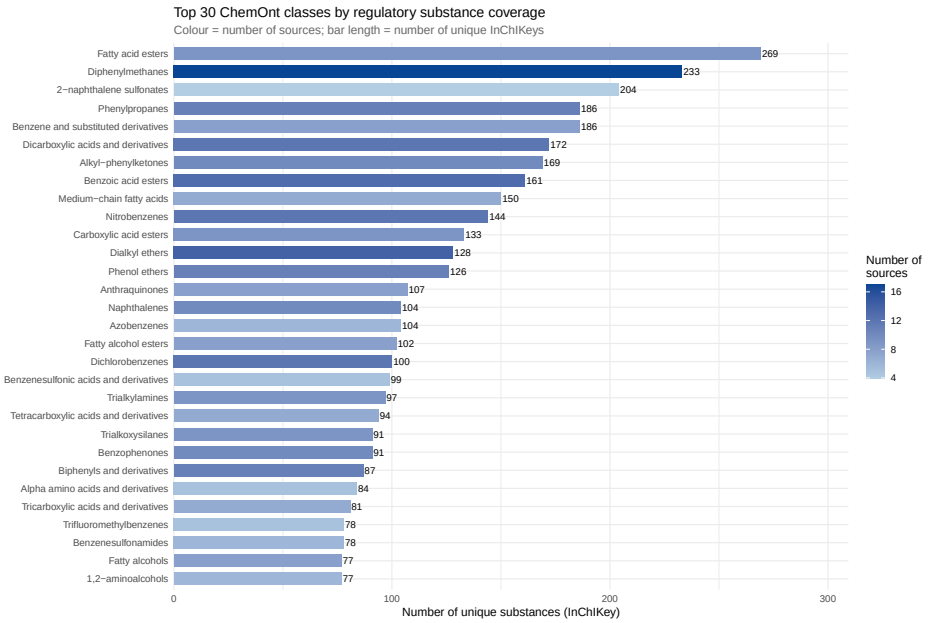

Figure 13. Top 30 ChemOnt direct-parent classes ranked by number of unique regulatory substances (InChIKeys) and number of distinct regulatory sources in which they occur.

Figure 14 reveals the long-tail character of ChemOnt class coverage: a small number of broad classes account for a disproportionately large share of regulated substances, while the majority of ChemOnt classes are represented by only a handful of entries. This concentration pattern has practical implications for class-level regulatory prioritisation: interventions targeting the most prevalent ChemOnt classes would cover a substantial fraction of the regulated chemical space.

2.9. Knowledge Graph Construction

All processed data — linkability taxonomy assignments, k-means cluster labels, ClassyFire direct-parent annotations, and ChEBI role annotations — were serialised to a unified RDF Turtle knowledge graph as described in Section 1.9. The graph integrates substance-level nodes (one per unique InChIKey or substance name), their regulatory source memberships, chemical taxonomy classifications, and biological role annotations. Semantic enrichment via SPARQL CONSTRUCT queries over the merged ChemOnt 2.1 SKOS ontology and ChEBI OWL graph enables downstream SPARQL-based queries across the regulatory, structural, and biological dimensions of the dataset. The output file `data/processed/rdf/substances_taxonomy.ttl` constitutes the machine-readable foundation for the priority scoring analysis described in the following subsection.

2.10. Workload Estimation for Manual Curation

Across the full combined ECHA dataset, 15,432 regulatory entries lack a resolvable chemical structure (Section 2.2). Establishing the pairwise semantic relationships — sub-

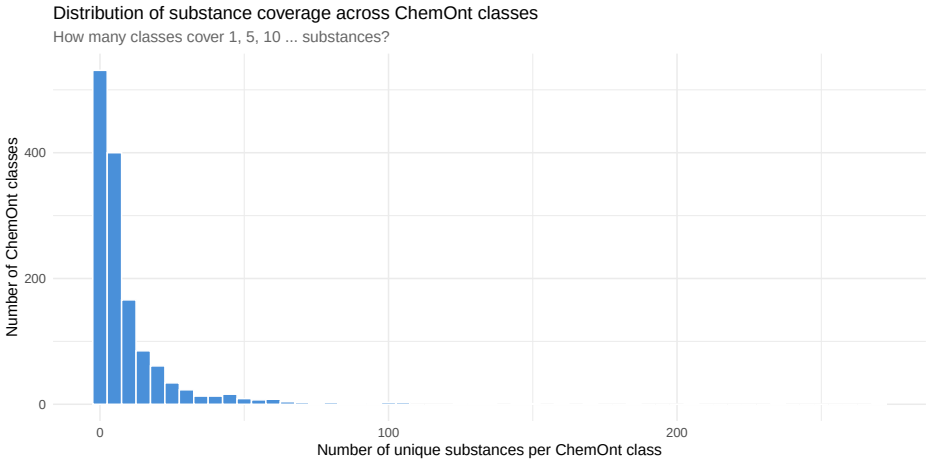

Figure 14. Distribution of the number of unique regulatory substances per ChemOnt class, illustrating the long-tail character of chemical class coverage in the combined ECHA regulatory dataset.

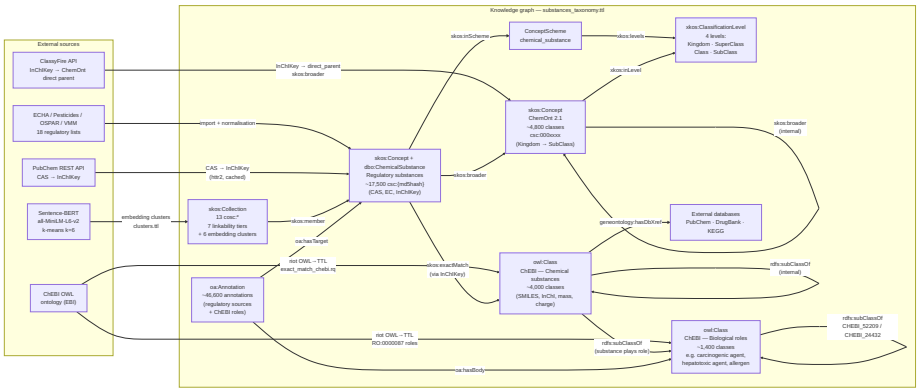

Figure 15. knowledge graph construction

set, overlap, or equivalence — among these entries requires expert assessment of every unique pair, yielding a total of

$$\binom{15{,}432}{2} = 119{,}055{,}096 \text{ pairs.}$$

At a throughput of 100 pair assessments per person-day, this corresponds to approximately 1,190,551 person-days — **nearly 6,000 person-years of continuous expert effort**. Figure 16 plots this workload across all regulatory sources, illustrating both the absolute scale and the quadratic growth that makes each additional non-structure entry disproportionately costly.

Manual pairwise curation at this scale is not operationally feasible. The only scalable alternative is to prevent the problem at the source: by requiring structure-resolvable identifiers and ontology-based representations in regulatory definitions wherever chemically possible, the population of entries requiring manual reconciliation can be structurally reduced rather than managed retrospectively.

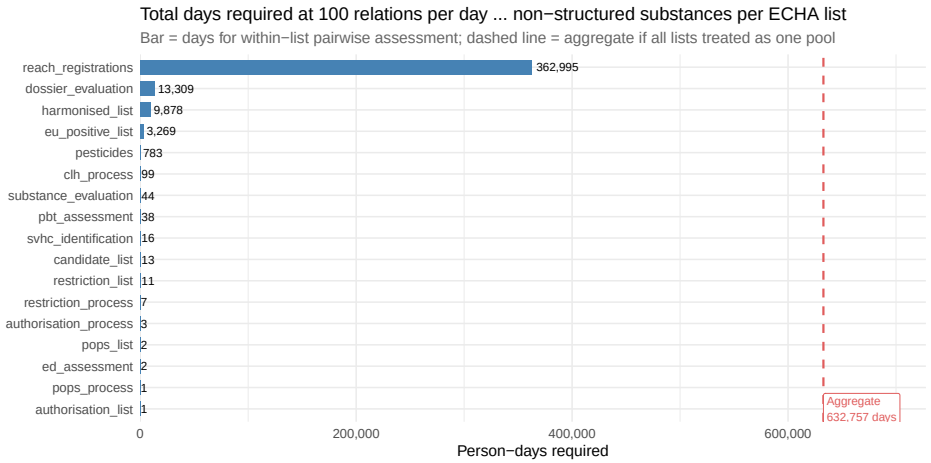

Figure 16. Pairwise curation workload for all 15,432 non-structure entries across the combined ECHA regulatory dataset, broken down by source. The total reaches approximately 1.2 million person-days — nearly 6,000 person-years — at an assumed throughput of 100 pair assessments per person-day. Manual curation at this scale is not operationally feasible; interoperability must be addressed at the legislative source.

2.11. Composite Priority Scoring

The composite priority scoring model (Section 1.11) was applied to 17,518 structure-defined substances with at least one regulatory list membership. Scores combine regulatory breadth (list_score) with biological hazard evidence from ChEBI (bio_score).

2.11.1. Top-Ranked Substances

Figure 17 presents the top 50 substances by priority score. Bisphenol A (2,2-bis(4-hydroxyphenyl)propane; BPA) achieves the highest score of 33 (list_score = 30, bio_score = 3), appearing across 12 distinct regulatory sources including the Candidate List, Restriction List, Authorisation Process, and Harmonised Classification List. Dibutyl phthalate ranks second with a score of 29 (list_score = 26, bio_score = 3), present in 10 sources. The third-ranked substance, 2-(2H-benzotriazol-2-yl)-4,6-di-tert-pentylphenol (score 27), achieves a high regulatory score through simultaneous listing in eight sources including the POPs list and Authorisation List, but receives no biological hazard score, indicating a gap in current ChEBI role annotations for this compound.

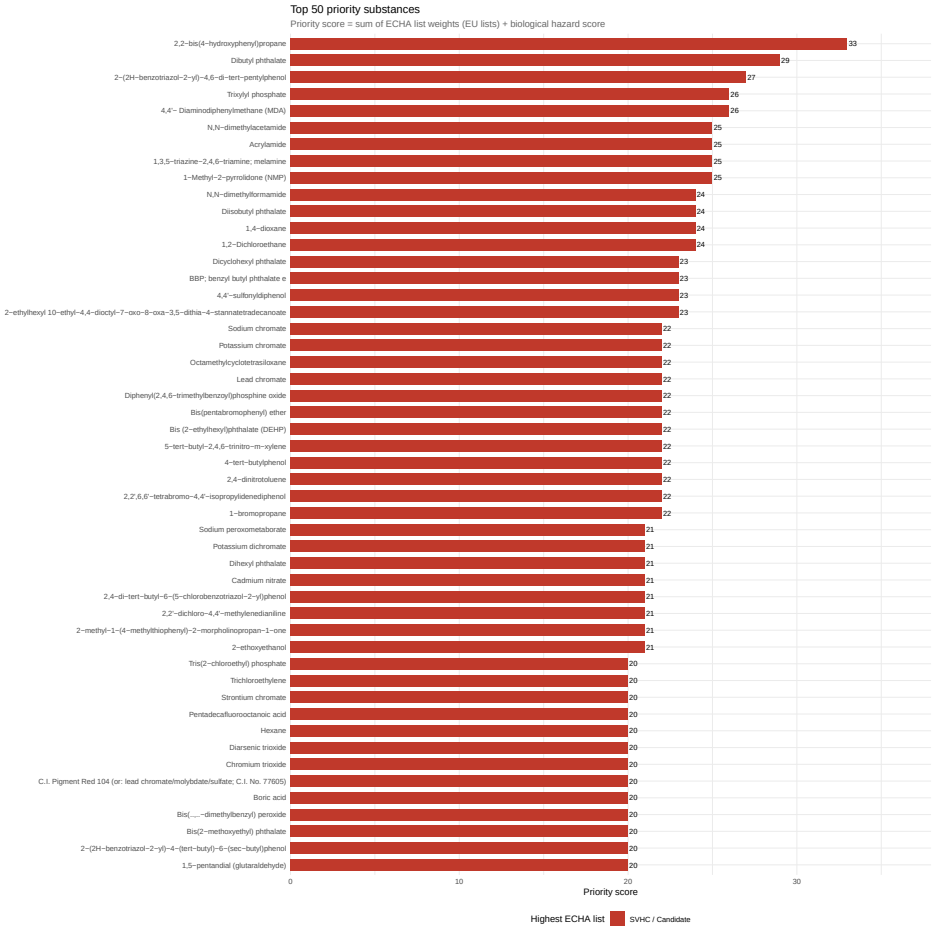

Figure 17. Top 50 structure-defined substances by composite priority score, showing the contribution of regulatory list score and biological hazard score. BPA achieves the maximum observed score of 33.

2.11.2. Regulatory List Co-occurrence

The regulatory list heatmap (Figure 18) visualises the co-occurrence of high-priority substances across the principal obligation lists. Substances appearing in both the Candidate List and the SVHC Identification List form the densest intersection; the Authorisation List and Restriction List constitute a secondary cluster. The heatmap confirms that the highest-priority substances are systematically distributed across multiple complementary regulatory instruments rather than being captured by any single list.

The UpSet plot in Figure 19 further decomposes these intersections, confirming that the largest multi-list intersections involve the Candidate List, SVHC Identification List, and Authorisation-related lists.

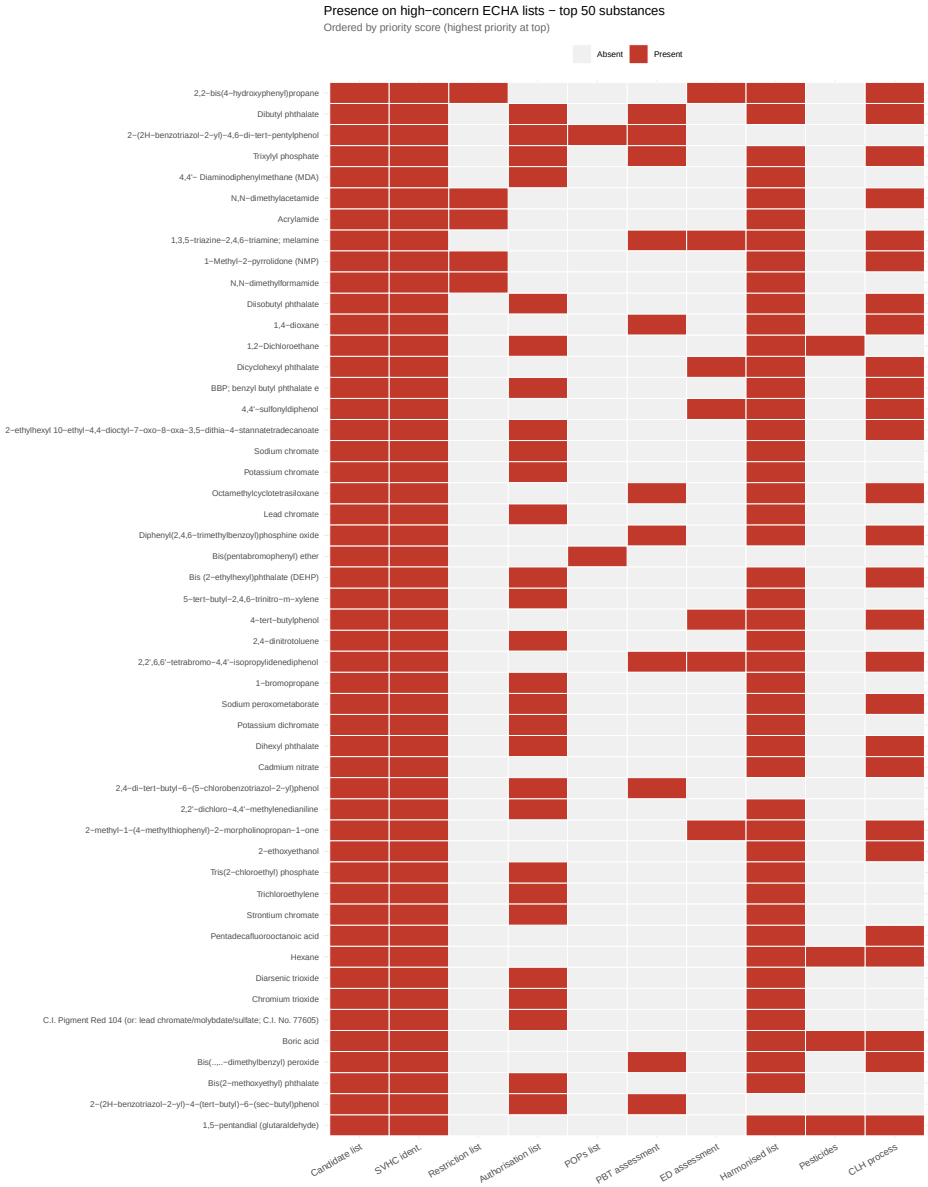

Figure 18. Heatmap of structure-defined substance co-occurrence across the principal obligation lists for the top-scoring priority substances. Substances at the intersection of multiple lists receive higher list scores.

2.11.3. Biological Hazard Profile

Of the 17,518 scored substances, 522 received a non-zero bio_score based on ChEBI role annotations. Figure 20 shows the distribution of ChEBI biological activity labels across these substances. Herbicidal activity is the most prevalent annotation (139 substances; 26.6% of scored entries), followed by allergen (76; 14.6%) and carcinogenic agent (67;

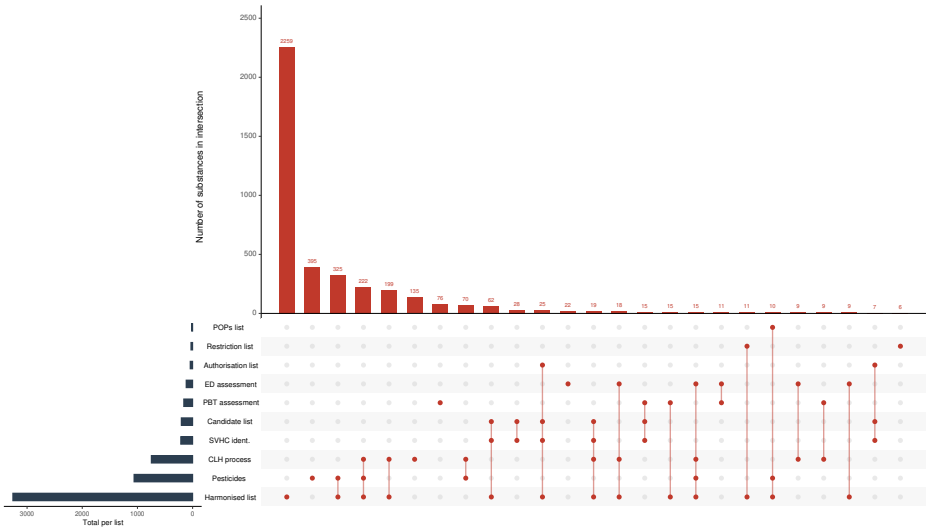

Figure 19. UpSet plot of high-priority substance membership across the principal obligation lists, ordered by intersection size. The Candidate List and SVHC Identification List account for the largest shared intersections.

12.8%). The predominance of herbicides reflects the large contribution of the pesticides database to the total dataset. Among the high-hazard-weight categories (weight 3; CMR and persistent environmental contaminants), carcinogenic agent (67 substances), mutagen (35), genotoxin (20), teratogenic agent (16), persistent organic pollutant (16), and endocrine disruptor (5) together account for 30.6% of biologically scored substances.

2.11.4. Priority Space Structure

The bubble chart (Figure 21) maps the full priority space, plotting list_score against bio_score for all scored substances. The upper-right quadrant (high list score, high bio score) identifies the substances of greatest combined regulatory and biological concern; BPA and Dibutyl phthalate occupy prominent positions in this quadrant.

The Sankey diagram (Figure 22) traces biological activity labels through to the regulatory lists in which the associated substances appear, revealing which obligation lists are most exposed to each hazard category. Carcinogens and mutagens flow predominantly into the Harmonised Classification List and Candidate List pathways, consistent with the CMR-based selection criteria of SVHC regulation. Herbicides and pesticide-related substances flow primarily through the pesticides database and, to a lesser extent, the Restriction List.

Finally, the treemap (Figure 23) partitions the biologically scored substances into six hazard groups, with tile area proportional to group size and tile colour encoding mean priority score. The POP/Environmental group (20 substances) exhibits the highest mean priority score (12.3), reflecting the co-occurrence of environmental persistence with broad regulatory listing. The Carcinogenic/Mutagenic group (108 substances, mean 9.6) is the largest high-hazard category by substance count. The Pesticidal group (245 substances) is the most numerous but carries a lower mean priority score (5.3), consistent with the lower per-list regulatory weights assigned to pesticide-specific lists.

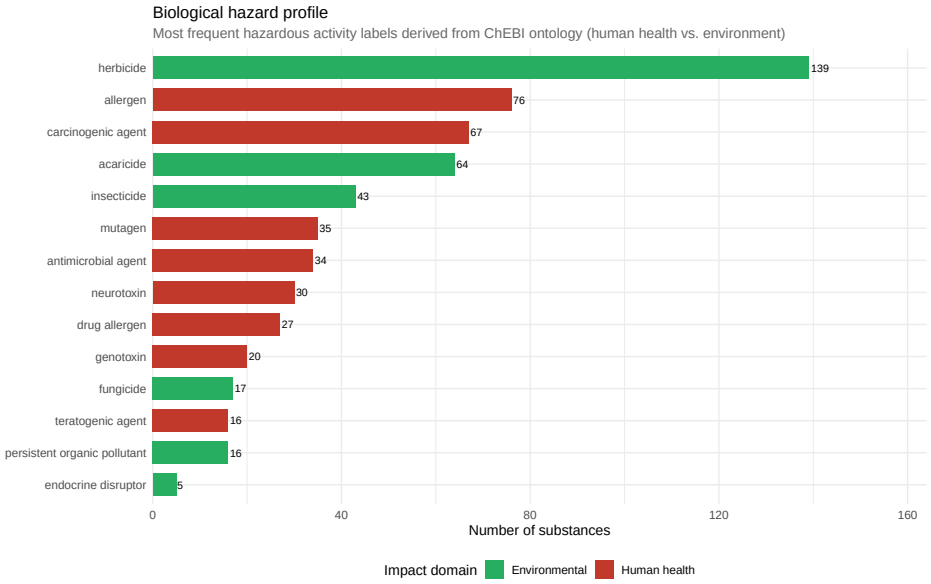

Figure 20. Distribution of ChEBI biological activity labels across the 522 substances with non-zero biological hazard scores. Labels are grouped by impact domain (human health vs. environmental) and hazard weight (1–3).

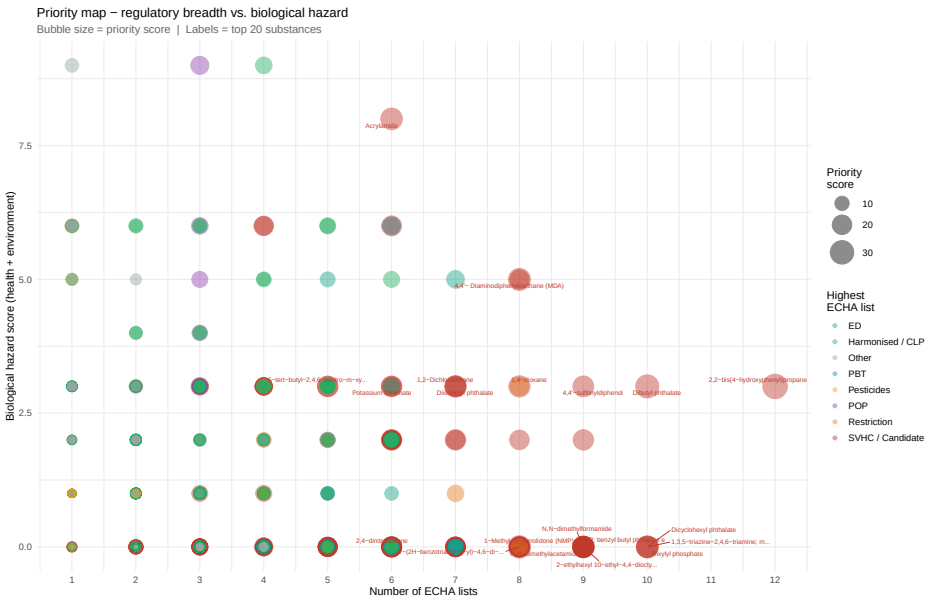

Figure 21. Bubble chart of composite priority space: regulatory list score versus biological hazard score for all scored substances. Bubble size is proportional to the number of regulatory source memberships. Substances in the upper-right quadrant combine broad regulatory coverage with documented biological hazard evidence.

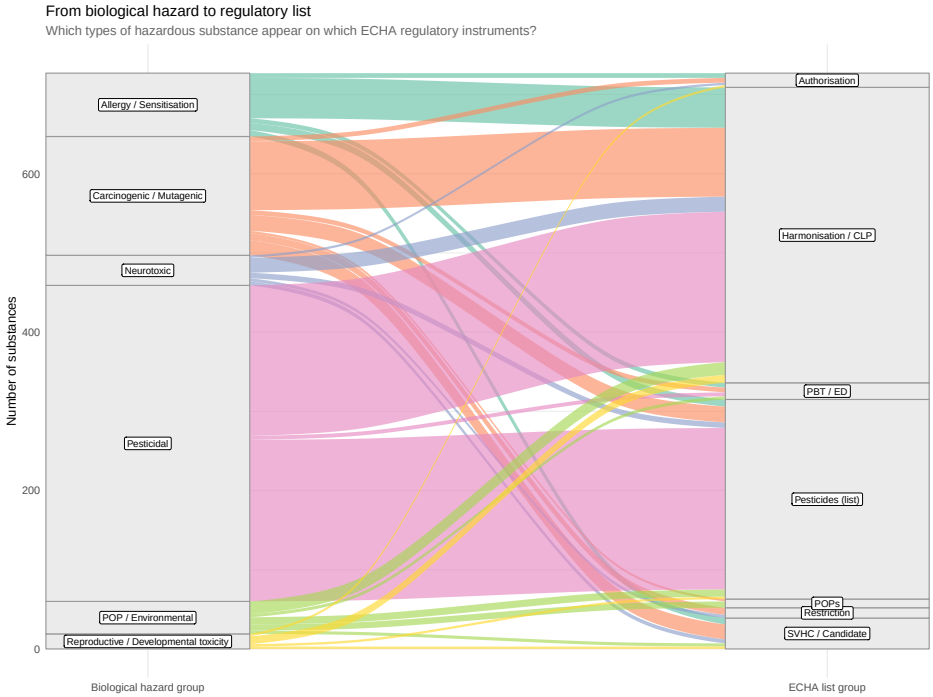

Figure 22. Sankey diagram linking ChEBI biological activity labels (left) to regulatory list memberships (right) for biologically scored substances. Flow width is proportional to the number of substances.

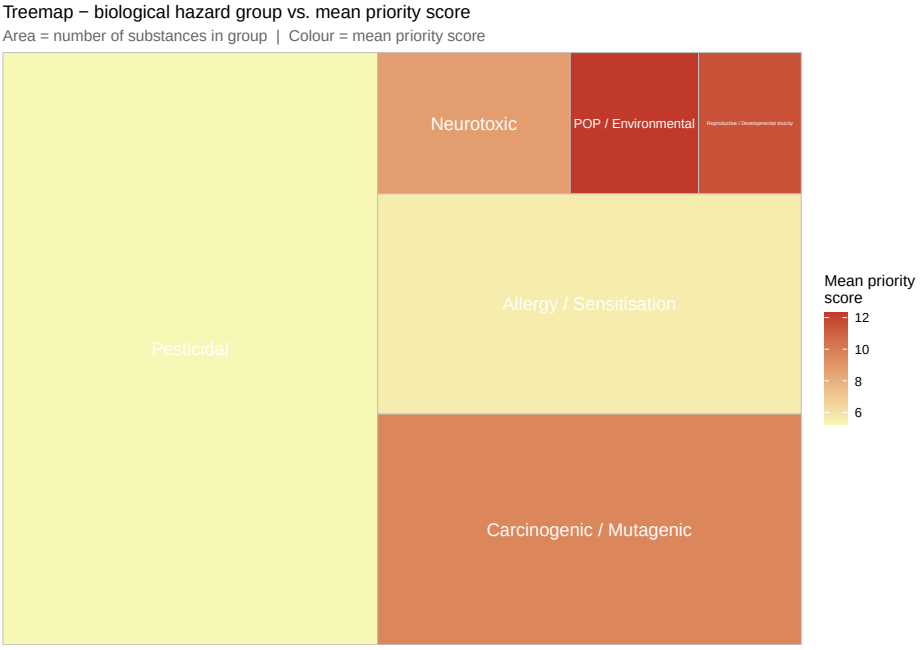

Figure 23. Treemap of biologically scored substances partitioned by hazard group. Tile area represents group size (number of substances); colour encodes mean composite priority score. The POP/Environmental group achieves the highest mean score despite being the smallest group.

References

[1] Kim S, Thiessen PA, Bolton EE, Bryant SH. PUG-SOAP and PUG-REST: web services for programmatic access to chemical information in PubChem. Nucleic acids research. 2015 Jul;43(W1):W605-11. Available from: https://doi.org/10.1093/nar/gkv396.

[2] R Core Team. R: A Language and Environment for Statistical Computing; 2024. R Foundation for Statistical Computing, Vienna, Austria. Available from: https://www.R-project.org/.

[3] Cleveland WS. Visualizing Data. Summit, NJ: Hobart Press; 1993. Available from: https://jwmason.org/wp-content/uploads/2021/08/Cleveland-1993-Visualizing-Data.pdf.

[4] Tufte ER. The Visual Display of Quantitative Information. 2nd ed. Cheshire, CT: Graphics Press; 2001. Available from: https://jwmason.org/wp-content/uploads/2021/08/Tufte-The-Visual-Display-of-Quantitative-Information.pdf.

[5] Lex A, Gehlenborg N, Strobelt H, Vuillemot R, Pfister H. UpSet: Visualization of Intersecting Sets. IEEE Transactions on Visualization and Computer Graphics. 2014;20(12):1983-92.

[6] Ward JH. Hierarchical Grouping to Optimize an Objective Function. Journal of the American Statistical Association. 1963;58(301):236-44.

[7] Murtagh F, Legendre P. Ward's Hierarchical Agglomerative Clustering Method: Which Algorithms Implement Ward's Criterion? Journal of Classification. 2014;31(3):274-95.

[8] Fruchterman TMJ, Reingold EM. Graph Drawing by Force-Directed Placement. Software: Practice and Experience. 1991;21(11):1129-64.

[9] Reimers N, Gurevych I. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics; 2019. p. 3982-92.

[10] Wang W, Wei F, Dong L, Bao H, Yang N, Zhou M. MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers. In: Advances in Neural Information Processing Systems. vol. 33; 2020. p. 5776-88. Available from: https://proceedings.neurips.cc/paper/2020/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html.

[11] Rousseeuw PJ. Silhouettes: A Graphical Aid to the Interpretation and Validation of Cluster Analysis. Journal of Computational and Applied Mathematics. 1987;20:53-65.

[12] Lloyd SP. Least Squares Quantization in PCM. IEEE Transactions on Information Theory. 1982;28(2):129-37.

[13] McInnes L, Healy J, Melville J. UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction. arXiv preprint arXiv:180203426. 2018. Available from: https://arxiv.org/abs/1802.03426.

[14] Manning CD, Raghavan P, Schütze H. Introduction to Information Retrieval. Cambridge: Cambridge University Press; 2008.

[15] Djoumbou Feunang Y, Eisner R, Knox C, Chepelev L, Hastings J, Owen G, et al. ClassyFire: automated chemical classification with a comprehensive, computable taxonomy. Journal of Cheminformatics. 2016 Nov;8(1):61. Available from: https://doi.org/10.1186/s13321-016-0174-y.

[16] Miles A, Bechhofer S. SKOS Simple Knowledge Organization System Reference. World Wide Web Consortium; 2009. Available from: https://www.w3.org/TR/skos-reference/.

[17] Sanderson R, Ciccarese P, Young B. Web Annotation Data Model. World Wide Web Consortium; 2017. Available from: https://www.w3.org/TR/annotation-model/.

[18] Reif DM, Martin MT, Tan SW, Houck KA, Judson RS, Richard AM, et al. Endocrine Profiling and Prioritization of Environmental Chemicals Using ToxCast Data. Environmental Health Perspectives. 2010;118(12):1714-20.

[19] European Parliament, Council of the European Union. Regulation (EC) No 1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH). Official Journal of the European Union; 2006. L 396. Available from: http://data.europa.eu/eli/reg/2006/1907/2022-03-01.

[20] European Parliament, Council of the European Union. Regulation (EC) No 1272/2008 on classification, labelling and packaging of substances and mixtures (CLP). Official Journal of the European Union; 2008. L 353. Available from: http://data.europa.eu/eli/reg/2008/1272/oj.

[21] European Parliament, Council of the European Union. Regulation (EU) 2019/1021 on persistent organic pollutants (POPs). Official Journal of the European Union; 2019. L 169. Available from: http://data.europa.eu/eli/reg/2019/1021/oj.

[22] European Commission. Chemicals Strategy for Sustainability Towards a Toxic-Free Environment. European Commission; 2020. COM(2020) 667 final. Available from: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52020DC0667.